higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATERIALS
NQF LEVEL 3

14 NOVEMBER 2019

This marking guideline consists of 6 pages.

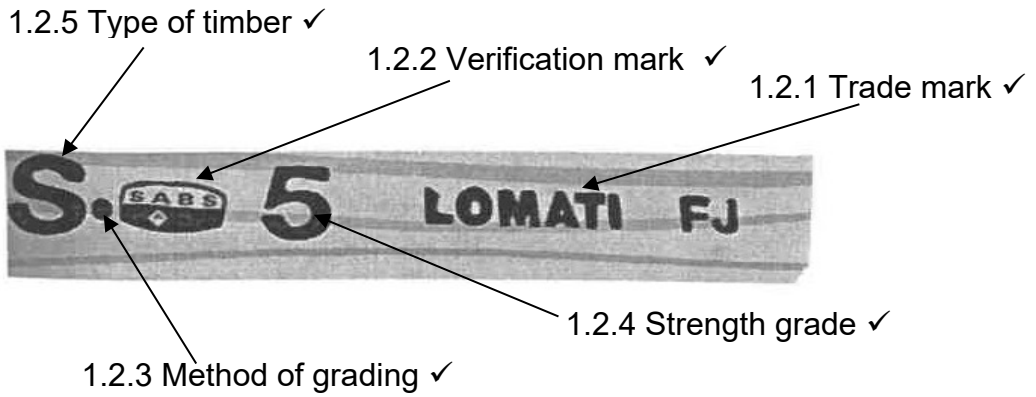
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QUESTION 1

1.1
1.1.1 False
1.1.2 True
1.1.3 False
1.1.4 False
1.1.5 True
1.1.6 False
1.1.7 False
1.1.8 True
1.1.9 False
1.1.10 True

(10 × 1) (10)

1.2



(5)

[15]

QUESTION 2

2.1
2.1.1 Sawing
2.1.2 Flame cutting
2.1.3 Cropping
2.1.4 Guillotining
2.1.5 Punching
2.1.6 Work-hardening
2.1.7 Annealing
2.1.8 Drawing
2.1.9 Plasma arc cutting
2.1.10 Laser cutting

(10 × 1) (10)

2.2
- Sawing✓ – Use a blade to cut through the metal plate, using a hand saw, circular saw or band saw.✓
- Flame cutting✓– Burn a slot into a sheet of ferrous metal using pure oxygen.✓
- Cropping✓ – Cut the ends of pieces of metal that have defects in them.✓
- Guillotining – A large machine with a blade that cuts metal sheets to specified sizes.
- Laser cutting – Laser beam is used to cut through metal.
- Abrasive water-jetting – Use high-pressure water to cut through the metal.
- Plasma cutting – Use gasses that conduct electricity to transfer heat energy from the power source to the cutting torch. (Any 3 × 2) (6)

2.3 Pure metal is a metal that is not mixed with any other metals.✓
Examples
- Aluminium, iron, copper, lead, zinc, tin.✓
- An alloy is mixture of metals.✓

Examples
Brass, steel, stainless steel, bronze, cast iron.✓ (4)
[20]

QUESTION 3

3.1 Mild steel contains low amounts of carbon, which makes it stronger and stiffer than other types of steel.✓ Because of its magnetic properties, it is mostly used in cars and electrical appliances.✓ High-tensile steel is characterised by a great amount of tensile strength and high yield point,✓ which renders it a preferred steel material in the construction industry.✓ (4)

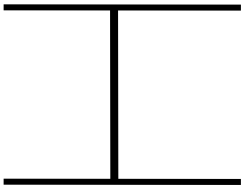
3.2 3.2.1 A stage where steel is no longer able to handle stress/pressure and starts to deform permanently.

3.2.2 A point where elastic deformation reaches its maximum strength and if the force applied increases, the steel will eventually break.

3.2.3 A stage where the stress is proportional to strain and if the force is removed, the material will go back to its original shape.

(3 × 2) (6)

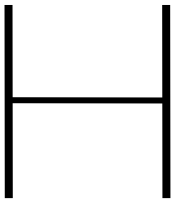
3.3 3.3.1



3.3.2



3.3.3 H



(3 × 2) (6)

3.4 The handing-over certificate is issued after the final inspection to show that the work has been carried out as per specification,✓ the work has been completed as per contractual agreement,✓ the area has been left in a safe condition✓ and any defective work has been fixed.✓ (4)

[20]

QUESTION 4

4.1 4.1.1 Drainage design: when the structure itself is designed in such a way that the water flows away from the building.

4.1.2 Painting: protecting metal against corrosion by creating a barrier between material and environment by means of paint.

4.1.3 Hot-dip galvanising: steel is submerged into a molten bath of zinc, to be coated with the thin layer of zinc.

4.1.4 Electroplating: an object is coated with a thin layer of metal using electricity.

4.1.5 Inherent corrosion resistance: refers to materials such as gold, which are naturally resistant to corrosion due to their fundamental make-up.

(5 × 2) (10)

4.2
- Ferritic stainless steels
- Austenitic stainless steels
- Weathering stainless steels
- Duplex stainless steels
- Martensitic stainless steels (Any 3 × 1) (3)

4.3
- Drainage design
- Painting
- Hot-dip galvanising (Any 2 × 1) (2)

[15]

QUESTION 5

5.1
5.1.1 American Welding Society
5.1.2 Welding procedure specification
5.1.3 Shielded metal arc welding
5.1.4 Flux-cored arc welding
5.1.5 Submerged arc welding
5.1.6 Manual metal arc welding

(6 × 1) (6)

5.2
- Strong and tight joining.
- General welding equipment is not very costly.
- Portable welding equipment is available.
- Welding permits considerable freedom in design.
- Various similar and dissimilar metals and alloys can be joined by welding.
- The parts that are welded together do not have to fit exactly. (Any 5 × 1) (5)

5.3 To ensure that proper air flow is provided throughout the welding environment to control and get rid of harmful contaminants. (2)

5.4 Soldering is done at a lower temperature, using either a propane torch or an electric heater,✓ while brazing is done at a higher temperature with oxyacetylene or map gas torches.✓ (2)

5.5
- Gas welding
- Friction welding
- Spot welding
- Submerged arc welding
- Tungsten inert gas welding
- Flux-cored arc welding (Any 5 × 1) (5)

[20]

QUESTION 6

6.1 Glass that is used for internal frameless shower doors and office partitioning is identified as toughened glass, OR glass that shatters into many tiny pieces when it is broken, is identified as toughened glass.

6.2 Glass with two or more layers that has an ability to block UV light and used as the windshields for cars and front shops is identified as laminated glass.

6.3 Glass that can hold its pieces together when broken and is used as a cost-effecive fire glass in public buildings, is identified as wired glass.

6.4 The most affordable glass that is used for domestic housing as window panes is identified as float glass.

6.5 Glass that can be manufactured in different colours and be mounted in wood, PVC and aluminium is identified as glazed panel.

(5 × 2) **[10]**

TOTAL: 100

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